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WORK VEHICLE

Abstract

A work vehicle includes a gear case in which lubricating oil is stored, and a transmission case positioned rearward of the gear case and in which lubricating oil is stored. The work vehicle also includes a communication passage to allow lubricating oil to flow back and forth between the gear case and the transmission case. A rear port, which is in communication with an interior of the transmission case and is for connection to a communication passage, is lower than a front port that is in communication with the interior of the gear case and is for connection to the communication passage.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to Japanese Patent Application No. 2022-210897 filed on Dec. 27, 2022 and is a Continuation Application of PCT Application No. PCT/JP2023/031281 filed on Aug. 29, 2023. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to work vehicles.

2. Description of the Related Art

[0003] As disclosed in JP 2013-141955A, an electric work vehicle, which is one example of a work vehicle, includes a motor (corresponding to a motor section) and a gear case, and further includes a transmission case provided rearward of the gear case.

[0004] In JP 2013-141955A, motive power output from the motor is transmitted to a PTO transmission (corresponding to a power receiver) via a gear transmission housed in the gear case, and then transmitted from the PTO transmission to a PTO shaft. The motive power output from the motor is transmitted to a transmission housed in the transmission case, and then transmitted from the transmission to front wheels and rear wheels (corresponding to travel devices).

SUMMARY OF THE INVENTION

[0005] In JP 2013-141955A, lubricating oil is stored inside the gear case, lubricating oil is also stored inside the transmission case, and the gear case and the transmission case serve as oil baths.

[0006] In this case, the gear case and the transmission case may be connected via a communication passage, and lubricating oil may travel back and forth between the gear case and the transmission case via the communication passage. This promotes the dissipation of heat from the lubricating oil.

[0007] Example embodiments of the present invention lower the overall center of gravity of work vehicles including communication passages that allow lubricating oil to flow back and forth between gear cases and transmission cases.

[0008] A work vehicle according to an example embodiment of the present invention includes a motor, a travel device, a gear case that houses a gear transmission configured to receive motive power from the motor and is configured to store lubricating oil, a power receiver configured to receive motive power from the gear transmission, a transmission case positioned rearward of the gear case to house a transmission to transmit motive power from the motor to the travel device and is configured to store lubricating oil, a communication passage configured to allow lubricating oil to flow back and forth between the gear case and the transmission case, a rear port in communication with an interior of the transmission case and connected to the communication passage, and a front port in communication with an interior of the gear case and connected to the communication passage, the rear port being lower than the front port.

[0009] According to this example embodiment of the present invention, the work vehicle includes the gear case, the transmission case positioned rearward of the gear case, the communication passage connecting the gear case to the transmission case, the rear port that is in communication with the interior of the transmission case and is for connection to the communication passage, and the front port that is in communication with the interior of the gear case and is for connection to the

communication passage, and in this example embodiment, the rear port is lower than the front port, and therefore the transmission case is able to be located at a lower position in the work vehicle. [0010] By making it easier for the transmission case to be at a lower position in the work vehicle, the lower position of the transmission case in the work vehicle can lower the overall center of gravity of the work vehicle, thus making it possible to improve the stability of the work vehicle during traveling.

[0011] In an example embodiment of the present invention, it is preferable that the work vehicle further includes a battery to supply electric power to the motor, and the gear case is positioned rearward of the motor.

[0012] According to this example embodiment of the present invention, the gear case is positioned rearward of the motor, and therefore when the communication passage is connected to the front port and the rear port, the motor has less influence on the arrangement of the communication passage.

[0013] This allows the communication passage to be connected to the front port and the rear port with less influence from the motor, thus making it possible to improve the ease of attachment of the communication passage and improve productivity.

[0014] In an example embodiment of the present invention, it is preferable that the power receiver includes a hydraulic pump, and the work vehicle further includes a supply passage configured to supply lubricating oil from the gear case and the transmission case to the hydraulic pump as hydraulic oil.

[0015] According to this example embodiment of the present invention, in the case where the power receiver includes a hydraulic pump, the lubricating oil in the gear case and the transmission case is supplied to the hydraulic pump as hydraulic oil via the oil supply passage.

[0016] In this case, the gear case and the transmission case serve as a hydraulic oil tank with a large volume that is a combination of the volumes of the gear case and the transmission case, thus making it unlikely to have a shortage of hydraulic oil in a configuration in which hydraulic oil is supplied from the hydraulic pump to a hydraulic device or the like.

[0017] In an example embodiment of the present invention, it is preferable that the work vehicle further includes a continuously variable transmission in a front portion of the transmission case to change a speed of motive power received from the motor and transmit the motive power to the transmission, and a continuously variable transmission case which is in a front portion of the transmission case to house the continuously variable transmission, the rear port is provided in the continuously variable transmission case, and the work vehicle further includes an internal passage provided in the continuously variable transmission case and connecting the rear port to the interior of the transmission case.

[0018] According to this example embodiment of the present invention, the continuously variable transmission case is provided in a front portion of the transmission case, and the continuously variable transmission is housed in the continuously variable transmission case. Motive power output from the motor section is transmitted to the continuously variable transmission, the speed of the motive power is changed, and the resulting motive power is then transmitted to the transmission in the transmission case and then transmitted from the transmission to a travel device.

[0019] Belt-type continuously variable transmissions are not oil-bathed, and hydrostatic continuously variable transmissions need to be configured such that the lubricating oil in the continuously variable transmission and the lubricating oil in the transmission case do not mix. Thus, the continuously variable transmission case and the transmission case are separated by a partition wall or the like.

[0020] According to the above example embodiment of the present invention, the rear port, to which the communication passage is connected, is provided in the continuously variable transmission case, and the internal passage that connects the rear port to the interior of the transmission case is provided in the continuously variable transmission case. As a result,

lubricating oil flows back and forth between the gear case and the transmission case via the communication passage and the internal passage, and lubricating oil in the gear case and the transmission case does not enter the continuously variable transmission case.

[0021] According to the above example embodiment of the present invention, by providing the internal passage in the continuously variable transmission case, it is not necessary to adopt a configuration in which the communication passage is arranged so as to bypass the continuously variable transmission case and is connected to the rear port of the transmission case, which is advantageous in terms of simplifying the structure.

[0022] In an example embodiment of the present invention, it is preferable that the communication passage includes a bent metal tube.

[0023] According to this example embodiment of the present invention, the communication passage includes a bent metal tube, and therefore the communication passage can be easily connected to the front port and the rear port even in a configuration in which the rear port is at a lower position than the front port, and the transmission case is at a lower position in the body.

[0024] In an example embodiment of the present invention, it is preferable that the work vehicle further includes a posture maintaining portion configured to maintain the communication passage in a posture in which a front portion of the communication passage is higher than a rear portion of the communication passage, and the posture maintaining portion is located at a connection between the communication passage and the rear port.

[0025] When the communication passage is connected to the rear port and the positional relationship between the gear case and the transmission case is fixed while the front portion of the communication passage is connected to the front port, the communication passage is in a posture in which the front portion is higher than the rear portion, and therefore, due to the weight of the communication passage, the communication passage may rotate relative to the rear port around an axis extending in the longitudinal direction of the communication passage, such that the front portion of the communication passage becomes lower than the rear portion.

[0026] In such a case, it is necessary to interrupt the process of fixing the positional relationship between the gear case and the transmission case and return the communication passage to a posture in which the front portion is higher than the rear portion.

[0027] According to the above example embodiment of the present invention, the posture maintaining portion, which is provided to maintain the communication passage in a posture in which the front portion is higher than the rear portion, is provided at the connection between the communication passage and the rear port.

[0028] As a result, when the communication passage is connected to the rear port and the positional relationship between the gear case and the transmission case is fixed while the front portion of the communication passage is connected to the front port, the communication passage is prevented from assuming a posture in which the front portion is lower than the rear portion, and therefore the process of fixing the positional relationship between the gear case and the transmission case can be performed with fewer interruptions, thus improving productivity.

[0029] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1 is a left side view of a tractor.

[0031] FIG. 2 is a schematic diagram illustrating a transmission system for transmission from a motor to front wheels and rear wheels, and a hydraulic system realized by a hydraulic pump.

[0032] FIG. **3** is a left side view of a region including the motor, a gear case, main frames, coupling frames, and a continuously variable transmission case.

[0033] FIG. **4** is a vertical sectional left side view of a region including the motor, the gear case, the main frames, the coupling frames, and the continuously variable transmission case.

[0034] FIG. **5** is a transverse sectional plan view of a region including the motor, the gear case, the main frames, the coupling frames, and the continuously variable transmission case.

[0035] FIG. **6** is a vertical sectional front view of a region including the main frames, the coupling frames, and the continuously variable transmission case.

[0036] FIG. **7** is a vertical sectional rear view of a region including the gear case, the main frames, and the coupling frames.

[0037] FIG. **8** is a plan view of the gear case and the hydraulic pump.

[0038] FIG. **9** is a rear view of the gear case.

[0039] FIG. **10** is a transverse sectional plan view of the gear case.

[0040] FIG. **11** is a left side view of a communication passage, a front port, and a rear port.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0041] FIGS. **1** to **11** show a tractor, which is an example of an electric work vehicle, and in FIGS. **1** to **11**, F indicates the forward direction, B indicates the rearward direction, U indicates the upward direction, D indicates the downward direction, R indicates the rightward direction, and L indicates the leftward direction.

[0042] As shown in FIGS. **1** and **2**, a body **3** is supported by right and left front wheels **1** (corresponding to travel devices) and right and left rear wheels **2** (corresponding to travel devices). A bonnet **4** is provided in a front portion of the body **3**, and a driver section **5** is provided in a rear portion of the body **3**. The driver section **5** includes a steering wheel **6** for steering the front wheels **1**, a driver seat **7**, a floor section **8**, and a ROPS frame **9**.

[0043] A battery **13**, an inverter **14**, and a motor **15** are provided in a front portion of the body **3**. Power stored in the battery **13** is supplied to the inverter **14**, and the DC power from the battery **13** is converted to AC power in the inverter **14** and supplied to the motor **15** to drive the motor **15**.

[0044] As shown in FIGS. **1**, **3** and **4**, the body **3** includes, for example, right main frames **10**, **16**, and **17**, left main frames **10**, **16**, and **17**, coupling frames **18** and **19**, a gear case **20**, a continuously variable transmission case **11**, and a transmission case **12**. A front axle case **21** is supported by front portions of the main frames **10**, and the front wheels **1** are held by the front axle case **21**. The rear wheels **2** are held by the transmission case **12**.

[0045] The right main frame **16** is coupled to a rear portion of the right main frame **10**. Furthermore, the left main frame **16** is coupled to a rear portion of the left main frame **10**. The right main frame **17** is coupled to a rear portion of the right main frame **16**. The left main frame **17** is coupled to a rear portion of the left main frame **16**. The continuously variable transmission case **11** is coupled to a front portion of the transmission case **12**, and the right and left main frames **17** are coupled to the continuously variable transmission case **11** and the transmission case **12**.

[0046] The gear case **20** couples the right main frames **10**, **16**, and **17** to the left main frames **10**, **16**, and **17** in the left-right direction, and the motor **15** is coupled to the gear case **20**. The battery **13** and the inverter **14** are supported by the main frame **10**, and the battery **13** is covered by the bonnet **4**.

[0047] As shown in FIGS. **5**, **6** and **7**, the coupling frame **18** has an upper surface portion **18a**, a right side surface portion **18b**, and a left side surface portion **18b**, and is channel-shaped (tunnel-shaped). Front portions of the right side surface portion **18b** and the left side surface portion **18b** of the coupling frame **18** are coupled to the gear case **20**, and rear portions of the right side surface portion **18b** and the left side surface portion **18b** of the coupling frame **18** are coupled to the continuously variable transmission case **11**.

[0048] The coupling frame **19** is shaped as a flat plate, and is below the coupling frame **18**. A front portion of the coupling frame **19** is coupled to the gear case **20**, and a rear portion of the coupling

frame **19** is coupled to the continuously variable transmission case **11**. A right portion of the coupling frame **19** is fastened to the right main frames **16** and **17** and the right side surface portion **18b** of the coupling frame **18**. Also, a left portion of the coupling frame **19** is fastened to the left main frames **16** and **17** and the left side surface portion **18b** of the coupling frame **18**.

[0049] As shown in FIGS. **7**, **8**, and **9**, the gear case **20** includes a first portion **20a** and a second portion **20b**, the first portion **20a** and the second portion **20b** are coupled to each other, and right and left coupling portions **20c** and **20d** are included in the second portion **20b**.

[0050] As shown in FIGS. **4** to **7**, a front portion of the coupling frame **19** is coupled to the coupling portion **20c** of the gear case **20**. An upper portion of the right side surface portion **18b** of the coupling frame **18** is coupled to an upper portion of the right coupling portion **20d** of the gear case **20**. Also, an upper portion of the left side surface portion **18b** of the coupling frame **18** is coupled to an upper portion of the left coupling portion **20d** of the gear case **20**. A lower portion of the right side surface portion **18b** of the coupling frame **18**, a lower portion of the right coupling portion **20d** of the gear case **20**, and the right main frames **16** and **17** are fastened together. Also, a lower portion of the left side surface portion **18b** of the coupling frame **18**, a lower portion of the left coupling portion **20d** of the gear case **20**, and the left main frames **16** and **17** are fastened together.

[0051] As a result, the gear case **20** is arranged extending in the left-right direction and is coupled to the right main frames **10**, **16**, and **17** and the left main frames **10**, **16**, and **17** in such a manner that the first portion **20a** of the gear case **20** is located on the front side and the second portion **20b** of the gear case **20** is located on the rear side.

[0052] As shown in FIGS. **4**, **5**, and **7**, the motor **15** is coupled to the first portion **20a** of the gear case **20**. As shown in FIGS. **7** and **10**, an output shaft **15a** of the motor **15** is inserted into the gear case **20**, and inside the gear case **20**, a transmission gear **32** (corresponding to a gear transmission) is coupled to the output shaft **15a** of the motor **15**. Transmission gears **33**, **34**, **35**, and **36** (corresponding to gear transmissions) are provided inside the gear case **20**, and adjacent gears among the transmission gears **32** to **36** are meshed with each other.

[0053] As shown in FIGS. **8** and **10**, hydraulic pumps **37** and **38** (corresponding to power receivers) are coupled to the first portion **20a** of the gear case **20**, and input shafts of the hydraulic pumps **37** and **38** are connected to the transmission gear **36**. Lubricating oil is stored inside the gear case **20**, and the interior of the gear case **20** serves as an oil bath.

[0054] As shown in FIG. **2**, a hydrostatic continuously variable transmission **24** is housed inside the continuously variable transmission case **11**, and a transmission shaft **25** extends between the output shaft **15a** of the motor **15** and the input shaft **24a** of the continuously variable transmission **24**.

[0055] Motive power output from the motor **15** is transmitted from the output shaft **15a** of the motor **15** to the transmission gears **32** to **36**, and the motive power output from the motor **15** is transmitted from the transmission gears **32** to **36** to the hydraulic pumps **37** and **38** to drive the hydraulic pumps **37** and **38**.

[0056] Motive power output from the motor **15** is also transmitted from the output shaft **15a** of the motor **15** to the continuously variable transmission **24** via the transmission shaft **25**. The continuously variable transmission **24** is capable of continuously changing the speed to both faster and slower speeds, and is operated by a speed change pedal **26** (see FIG. **1**) provided on the floor section **8**.

[0057] An auxiliary transmission **27** (corresponding to a transmission), a rear wheel differential device **28** (corresponding to a transmission), a front wheel transmission **29** (corresponding to a transmission), and right and left brakes **30** are housed inside the transmission case **12**. Motive power is transmitted from the continuously variable transmission **24** to the auxiliary transmission **27**, and then transmitted from the auxiliary transmission **27** to the rear wheels **2** via the rear wheel differential device **28**. Lubricating oil is stored inside the transmission case **12**, and the interior of the transmission case **12** serves as an oil bath.

[0058] According to the above-described configuration, the tractor includes the gear case **20** that houses the transmission gears **32** to **36** (gear transmissions), which receive motive power from the motor **15** (motor section), and that stores lubricating oil. The tractor further includes the hydraulic pumps **37** and **38** (driven sections) that receive motive power from the transmission gears **32** to **36** (gear transmissions).

[0059] The motor **15** and the hydraulic pumps **37** and **38** are provided in front of the gear case **20**, and the gear case **20** is provided behind the motor **15**.

[0060] The continuously variable transmission case **11** is provided in a front portion of the transmission case **12** and houses the continuously variable transmission **24** that receives motive power from the motor **15**, changes the speed of the motive power from the motor **15**, and transmits the resulting motive power to the auxiliary transmission **27** (transmission), the rear wheel differential device **28** (transmission), and the front wheel transmission **29** (transmission).

[0061] In this case, the interior of the continuously variable transmission case **11** and the interior of the transmission case **12** are separated by a partition wall, and therefore the lubricating oil in the continuously variable transmission case **11** and the lubricating oil in the transmission case **12** do not mix with each other.

[0062] As shown in FIG. 2, motive power branched off from a point between the auxiliary transmission **27** and the rear wheel differential device **28** is transmitted to the front wheel transmission **29**. The output shaft **29a** of the front wheel transmission **29** passes through the continuously variable transmission case **11** and exits from the front portion of the continuously variable transmission case **11**, and motive power from the front wheel transmission **29** is transmitted from the output shaft **29a** of the front wheel transmission **29** via the transmission shaft **22** to the front wheel differential device **23** housed in the front axle case **21**, and then from the front wheel differential device **23** to the front wheels **1**.

[0063] As shown in FIGS. 4 to 7, the transmission shafts **22** and **25** are arranged extending in the front-rear direction in a tunnel-shaped space surrounded by the coupling frames **18** and **19**. The transmission shaft **22** passes through a gap between the gear case **20** and the front portion of the coupling frame **19**, extends obliquely downward and forward, and is connected to the front axle case **21** (front wheel differential device **23**).

[0064] As shown in FIG. 2, when the front wheels **1** are turned from a straight ahead position to a position within a range of right and left set angles, the front wheel transmission **29** drives the front wheels **1** and the rear wheels **2** at the same speed. When the front wheels **1** are turned to the right or left beyond the right and left set angles, the front wheel transmission **29** drives the front wheels **1** at a higher speed than the rear wheels **2**.

[0065] The right brake **30** is provided in the transmission system for transmission from the rear wheel differential device **28** to the right rear wheels **2**, and the left brake **30** is provided in the transmission system for transmission from the rear wheel differential device **28** to the left rear wheels **2**. A right brake pedal **31** capable of operating the right brake **30**, and a left brake pedal **31** capable of operating the left brake **30** are provided on a left-front portion of the floor section **8** (see FIG. 1).

[0066] In order to turn right (turn left), the right (left) brake pedal **31**, which is on the turning center side, is depressed to apply the right (left) brake **30**, thereby enabling a tight turn.

[0067] According to the above configuration, the front wheels **1** and the rear wheels **2** are driven by the motor **15**.

[0068] The tractor includes the transmission case **12** that is provided rearward of the gear case **20** and houses the auxiliary transmission **27** (transmission), the rear wheel differential device **28** (transmission), and the front wheel transmission **29** (transmission), and stores lubricating oil. The auxiliary transmission **27** transmits motive power output from the motor **15** to the front wheels **1** and the rear wheels **2**.

[0069] As shown in FIG. 2, an internal passage **39** that extends in the front-rear direction is

provided inside the continuously variable transmission case **11**. The internal passage **39** is in communication with the interior of the transmission case **12**. A communication passage **40** connects the gear case **20** and the continuously variable transmission case **11**, and the communication passage **40** is in communication with the internal passage **39**. Furthermore, the communication passage **40** is in communication with the interior of the gear case **20**.

[0070] Lubricating oil can flow back and forth between the interior of the gear case **20** and the interior of the transmission case **12** through the communication passage **40** and the internal passage **39**, and the lubricating oil in the gear case **20** and the transmission case **12** does not mix with the lubricating oil in the continuously variable transmission case **11**.

[0071] According to the above-described configuration, the tractor includes the communication passage **40** that allows lubricating oil to flow back and forth between the gear case **20** and the transmission case **12**. Due to the communication passage **40** and the internal passage **39**, the gear case **20** and the transmission case **12** define a single hydraulic oil tank.

[0072] A filter **41** is provided in the transmission case **12**, and a supply passage **42** connects the filter **41** to the hydraulic pumps **37** and **38**. Lubricating oil from the gear case **20** and the transmission case **12** passes through the filter **41** and the supply passage **42**, and is supplied to the hydraulic pumps **37** and **38** as hydraulic oil.

[0073] According to the above-described configuration, the tractor includes the supply passage **42** that supplies lubricating oil from the gear case **20** and the transmission case **12** to the hydraulic pumps **37** and **38** as hydraulic oil.

[0074] As shown in FIG. 2, a power steering device **43** is provided below the steering wheel **6**, and the power steering device **43** is operated by the steering wheel **6**. The hydraulic pump **38** and the power steering device **43** are connected to each other by an oil passage **44**, and the power steering device **43** and the transmission case **12** are connected to each other by an oil passage **45**.

[0075] A double-action steering cylinder **46** to steer the front wheels **1** is provided in the front axle case **21**, and the power steering device **43** and the steering cylinder **46** are connected by an oil passage **47**.

[0076] Hydraulic oil from the hydraulic pump **38** is supplied to the power steering device **43** via the oil passage **44**, and when the steering wheel **6** is operated, hydraulic oil is supplied from the power steering device **43** to the steering cylinder **46** via the oil passage **47**, and the front wheels **1** are steered by the steering cylinder **46**.

[0077] The hydraulic oil discharged from the steering cylinder **46** is returned to the transmission case **12** via the oil passage **47**, the power steering device **43**, and the oil passage **45**. A fully-hydraulic power steering system includes the power steering device **43** and the steering cylinder **46**.

[0078] As shown in FIG. 2, an external hydraulic unit **48** is attached to the outside of the transmission case **12**, and two pairs of hydraulic ports **49** are attached to a lower portion of the floor section **8**. The hydraulic pump **37** and the external hydraulic unit **48** are connected by an oil passage **50**, and the external hydraulic unit **48** and the hydraulic ports **49** are connected by oil passages **51**.

[0079] When a hydraulic device such as a front loader (not shown) is attached to the body **3**, the hydraulic device (e.g., front loader) and the hydraulic ports **49** are connected by oil passages (not shown).

[0080] Hydraulic oil from the hydraulic pump **37** is supplied to the external hydraulic unit **48** via the oil passage **50**, and is then supplied from the external hydraulic unit **48** to the hydraulic device via the oil passages **51** and the hydraulic ports **49** to drive the hydraulic device.

[0081] When an operating lever **48a** of the external hydraulic unit **48** is operated, the hydraulic oil supplied from the external hydraulic unit **48** to the hydraulic device is operated to operate the hydraulic device. The hydraulic oil discharged from the hydraulic device returns to the external hydraulic unit **48** via the hydraulic ports **49** and the oil passages **51**, and is returned from the external hydraulic unit **48** to the transmission case **12**.

[0082] As shown in FIG. 1, a top link 52 and right and left lower links 53 are supported in a rear portion of the transmission case 12, and a work device (not shown) such as a rotary tiller can be connected to the top link and the lower links 53.

[0083] Right and left lift arms 54 are provided in a rear portion of the transmission case 12, and the lift arms 54 are connected to the lower links 53 by linking rods 55. When the lift arms 54 are swung up and down, the work device is raised and lowered via the top link 52 and the lower links 53.

[0084] As shown in FIG. 2, lift cylinders 56 provided to swing the lift arms 54, and a control valve 57, are provided in the transmission case 12. In the external hydraulic unit 48, the oil passage 50 branches into a system leading to the hydraulic ports 49 and a system leading to the control valve 57. The system leading to the control valve 57 is connected to the control valve 57 by an oil passage 58, and the control valve 57 is connected to the lift cylinders 56 by an oil passage 59.

[0085] Hydraulic oil from the hydraulic pump 37 is supplied to the control valve 57 via the oil passages 50 and 58, and is then supplied from the control valve 57 to the lift cylinders 56 via the oil passage 59, and thus the lift arms 54 are swung up and down by the lift cylinders 56. The hydraulic oil discharged from the lift cylinders 56 returns to the control valve 57 via the oil passage 59, and is then returned from the control valve 57 to the transmission case 12.

[0086] As shown in FIGS. 3, 4, 5, 7 and 9, a front port 60 is provided facing rearward in the second portion 20b of the gear case 20. The front port 60 is provided in the second portion 20b of the gear case 20 at a position lower than the output shaft 15a of the motor 15 and the transmission gear 32, on one side of the output shaft 15a of the motor 15 and the transmission gear 32, namely the side opposite to the transmission gears 33 to 36.

[0087] As shown in FIGS. 3 to 6, a rear port 61 is provided facing forward in a front portion of the continuously variable transmission case 11, and the rear port 61 is in communication with the internal passage 39 (see FIG. 2).

[0088] The rear port 61 is provided in the front portion of the continuously variable transmission case 11, at a position lower than the input shaft 24a of the continuously variable transmission 24 and the output shaft 29a of the front wheel transmission 29, on one side of the output shaft 29a of the front wheel transmission 29, namely the side opposite to the input shaft 24a of the continuously variable transmission 24.

[0089] As shown in FIGS. 3 to 6, the front port 60 and the rear port 61 are provided in a tunnel-shaped space surrounded by the coupling frames 18 and 19. When the gear case 20, the continuously variable transmission case 11, and the transmission case 12 are coupled to the main frames 10, 16, and 17, the rear port 61 is at a lower position than the front port 60.

[0090] As shown in FIG. 11, the communication passage 40 includes a metal pipe bent into a crank shape. A flange portion 62 is coupled to the front portion 40a of the communication passage 40. A flange portion 63 (corresponding to a posture maintaining portion) is coupled to the rear portion 40b of the communication passage 40, and a protrusion 63a that protrudes radially outward from the flange portion 63 is provided at one location on the outer peripheral portion of the flange portion 63. A notched recess 61a (corresponding to a posture maintaining portion) is provided at one location on the outer peripheral portion of the front end portion of the rear port 61.

[0091] As shown in FIGS. 3 to 7, when the gear case 20, the continuously variable transmission case 11, and the transmission case 12 are coupled to the main frames 10, 16, and 17 and the coupling frames 18 and 19, the communication passage 40 is also connected to the front port 60 and the rear port 61.

[0092] In this case, before connecting the gear case 20, the continuously variable transmission case 11, and the transmission case 12 to the main frames 10, 16, and 17 and the coupling frames 18 and 19, the rear portion 40b of the communication passage 40 is inserted into the rear port 61, and the protrusion 63a of the flange portion 63 is inserted into the recess 61a of the rear port 61, as shown in FIGS. 5, 6, and 11. The communication passage 40 is held in the rear port 61 in a posture in

which the front portion **40a** of the communication passage **40** is higher than the rear portion **40b** of the communication passage **40**.

[0093] In the above-mentioned state, the gear case **20**, the continuously variable transmission case **11**, and the transmission case **12** are moved toward each other, and the front portion **40a** of the communication passage **40** is inserted into the front port **60**.

[0094] In this case, since the protrusion **63a** of the flange portion **63** has been inserted into the recess **61a** of the rear port **61**, the communication passage **40** rotates around an axis passing through the rear port **61** in the front-rear direction due to the weight of the front portion **40a** of the communication passage **40**, and the communication passage **40** does not take a posture in which the front portion **40a** is lower than the rear portion **40b** of the communication passage **40**.

[0095] After inserting the front portion **40a** of the communication passage **40** into the front port **60** as described above, the gear case **20**, the continuously variable transmission case **11** and the transmission case **12** need only be coupled to the main frames **10**, **16**, and **17** and the coupling frames **18** and **19**. As shown in FIGS. **4**, **7** and **11**, the flange portion **62** is coupled to an upper portion of the front port **60** using a bolt **64**, and thus, the communication passage **40** connects the front port **60** and the rear port **61** to each other.

[0096] According to the above-described configuration, the tractor includes the flange portion **63** and the recess **61a** that maintain the communication passage **40** in a posture in which the front portion **40a** is higher than the rear portion **40b**. The flange portion **63** and the recess **61a** are provided at the connection between the communication passage **40** and the rear port **61**.

First Alternative Example Embodiment of Invention

[0097] The continuously variable transmission case **11** and the continuously variable transmission **24** may be omitted.

[0098] According to this configuration, the rear port **61** is provided in a front portion of the transmission case **12**, and the internal passage **39** is not required.

Second Alternative Example Embodiment of Invention

[0099] The recess **61a** of the rear port **61** may be omitted, the rear portion **40b** of the communication passage **40** may be inserted into the rear port **61**, and the flange portion **63** may be coupled to the rear port **61** by a bolt (not shown, but corresponding to a posture maintaining portion). According to this configuration, the bolt **64** to connect the flange portion **62** to the front port **60** is not necessary.

Third Alternative Example Embodiment of Invention

[0100] The communication passage **40** may include a flexible tube such as a rubber hose.

[0101] According to this configuration, after coupling the gear case **20**, the continuously variable transmission case **11**, and the transmission case **12** to the main frames **10**, **16**, and **17** and the coupling frames **18** and **19**, the communication passage **40** can be connected to the front port **60** and the rear port **61**, thus eliminating the need for posture maintaining portions such as the flange portion **63** and the recess **61a** of the rear port **61**.

Fourth Alternative Example Embodiment of Invention

[0102] Instead of the hydraulic pumps **37** and **38**, a work device such as a grass cutting device (not shown) may be provided as the power receiver, and motive power output from the transmission gear **36** of the gear case **20** may be transmitted to the work device.

Fifth Alternative Example Embodiment of Invention

[0103] Instead of the rear wheels **2**, a crawler travel device (not shown) may be provided as the travel device.

[0104] Instead of the front wheels **1** and the rear wheels **2**, a crawler travel device (not shown) may be provided as the travel device.

[0105] Instead of the motor **15**, an engine (not shown) may be provided as the motor section.

[0106] Example embodiments of the present invention are applicable not only to tractors, but also to other work vehicles such as electric carts and transport vehicles, and construction work vehicles

such as wheel loaders.

[0107] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A work vehicle comprising: a motor; a travel device; a gear case that houses a gear transmission configured to receive motive power from the motor, and is configured to store lubricating oil; a power receiver configured to receive motive power from the gear transmission; a transmission case positioned rearward of the gear case to house a transmission to transmit motive power from the motor to the travel device, and is configured to store lubricating oil; a communication passage configured to allow lubricating oil to flow back and forth between the gear case and the transmission case; rear port in communication with an interior of the transmission case and connected to the communication passage; and a front port in communication with an interior of the gear case and connected to the communication passage; the rear port being lower than the front port.
 2. The work vehicle according to claim 1, further comprising: a battery to supply electric power to the motor; wherein the gear case is positioned rearward of the motor.
 3. The work vehicle according to claim 1, wherein the power receiver includes a hydraulic pump; and the work vehicle further comprises a supply passage to supply lubricating oil from the gear case and the transmission case to the hydraulic pump as hydraulic oil.
 4. The work vehicle according to claim 1, further comprising: a continuously variable transmission in a front portion of the transmission case to change a speed of motive power received from the motor and transmit the motive power to the transmission; and a continuously variable transmission case in a front portion of the transmission case to house the continuously variable transmission; wherein the rear port is provided in the continuously variable transmission case; and the work vehicle further comprises an internal passage provided in the continuously variable transmission case and connecting the rear port to the interior of the transmission case.
 5. The work vehicle according to claim 1, wherein the communication passage includes a bent metal tube.
 6. The work vehicle according to claim 1, further comprising: a posture maintaining portion configured to maintain the communication passage in a posture in which a front portion of the communication passage is higher than a rear portion of the communication passage; wherein the posture maintaining portion is located at a connection between the communication passage and the rear port.
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